**MULTIVARIABLE CALCULUS HT23 SHEET 1**
**Multiple planar integrals. Change of variables.**

**1.** Let $a, b > 0$. By swapping the order of double integrals of $e^{-xy}$ on the set $[0, \infty) \times [a, b]$ show that

$$\int_0^\infty \frac{e^{-ax} - e^{-bx}}{x} \, \mathrm{d}x = \ln\left(\frac{b}{a}\right).$$

**2.** Show that

$$\int_{y=0}^{y=1} \left( \int_{x=0}^{x=1} \frac{x-y}{(x+y)^3} \, \mathrm{d}x \right) \mathrm{d}y = -\frac{1}{2} \neq \frac{1}{2} = \int_{x=0}^{x=1} \left( \int_{y=0}^{y=1} \frac{x-y}{(x+y)^3} \, \mathrm{d}y \right) \mathrm{d}x.$$

**3.** Show that

$$\iint\limits_{[0,1]^2} |x - y| \, \mathrm{d}A = \frac{1}{3}.$$

Why does this integral represent the mean distance between two points chosen randomly and independently from the unit interval $[0, 1]$?

**4.** (i) The *moment of inertia* of a uniform disc $D$ of radius $a$ and mass $m$ about an axis vertically through its centre equals

$$I_0 = \iint\limits_D r^2 \, \rho \, \mathrm{d}A,$$

where $\rho$ is the density of the disc. Determine $\rho$ in terms of $a$ and $m$. Use polar co-ordinates to show that $I_0$ equals $ma^2/2$.

(ii) The vertical axis is moved from the centre of the disc to a point distance $R \leqslant a$ away. The moment of inertia now equals

$$I_R = \iint\limits_D d(r, \theta, R)^2 \, \rho \, \mathrm{d}A$$

where $d(r, \theta, R)$ is the distance of the point $(r, \theta)$ from the axis. Find $I_R$ and show that it is an increasing function of $R$ for $0 \leqslant R \leqslant a$.

**5.** Let

$$I = \iint\limits_{[0,1]^2} \frac{\mathrm{d}x \, \mathrm{d}y}{1 - xy}.$$

(i) By applying the binomial theorem to the integrand, show that

$$I = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

(ii) Set $x = u - v$ and $y = u + v$. Determine the Jacobian $\partial(x, y)/\partial(u, v)$. Show that

$$I = 4 \iint\limits_T \frac{\mathrm{d}u \, \mathrm{d}v}{1 - u^2 + v^2}$$

where $T$ is the triangle with vertices $(0, 0)$, $(1, 0)$ and $(1/2, 1/2)$.

(iii) By splitting the interval $0 \leqslant u \leqslant 1$ into halves, and using trigonometric substitutions and identities, show that $I = \pi^2/6$.

**6.** (Optional) (i) Let $a \geqslant 0$. By applying the method of Exercise 1 to $e^{-yx} \sin x$, show that

$$\int_0^\infty \left(1 - e^{-ax}\right) \frac{\sin x}{x} \, \mathrm{d}x = \tan^{-1} a.$$

Deduce that $\int_0^\infty \frac{\sin x}{x} \, \mathrm{d}x = \pi/2$.

(ii) What is $\int_0^\infty \frac{\sin cx}{x} \, \mathrm{d}x$ where $c$ is a real number?

(iii) Show that the integrals $\int_0^{n\pi} \left|\frac{\sin x}{x}\right| \, \mathrm{d}x$ increase without bound as $n$ becomes large.